

# The friction on a quantized vortex in a truncated Gross–Pitaevskii field follows the temperature, not the normal density

A pre-registered test of  $\alpha \propto \rho_n$  against the cutoff, and the correction of a published reading

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## Abstract

The mutual friction  $\alpha$  that drags a quantized vortex through the thermal component of a two-dimensional Bose gas is commonly taken to scale with the normal fraction  $\rho_n/\rho$ . In an earlier paper [1] we measured  $\alpha = 0.0062, 0.014, 0.02$  at three temperatures of the projected Gross–Pitaevskii equation at one cutoff and found  $\alpha \simeq 0.24 \rho_n/\rho$ ; we read the coefficient as a mode-averaged transport cross-section of the vortex, 1.4 healing lengths, and registered the question of whether it depends on the cutoff, with the Born phonon-scattering law as the rival ( $\propto k_c$ ). We now measure  $\alpha$  at two further cutoffs,  $k_c\xi = 2\pi/3$  and  $2\pi$ , with the same energy-conserving instrument, with its  $T = 0$  known answer reproduced at each cutoff to 0.3 %. The registered prediction of a cutoff-independent coefficient fails, and so does its rival: the coefficient  $c = \alpha/(\rho_n/\rho)$  is  $0.31 \pm 0.03, 0.232 \pm 0.014$  and  $0.102 \pm 0.011$  at  $k_c\xi = 2.1, 3.1, 6.3$ , falling by a factor 3.1 while  $\rho_n/(\rho T)$  rises by a factor of three to four. What does not move is  $\alpha$  itself:  $\alpha/T = 0.054 \pm 0.003$  in six arms ( $\chi^2 = 1.3$  for 5 degrees of freedom) over three cutoffs and  $T = 0.10\text{--}0.22$  ( $0.060 \pm 0.002$  with the regression estimator: the primary energy estimator is biased low by diffusion, a flaw of the registered synthetic gate that a second implementation exposed). The proportionality to  $\rho_n$  reported before was a degeneracy of a single cutoff, where  $\rho_n \propto T$ ; the mode-averaged cross-section is not a property of the vortex (1.9, 1.4, 0.59 healing lengths) and the “geometric core size” reading is withdrawn. The temperature law was found after the data were read; it was registered as a hypothesis, with its prediction for the next arm fixed in advance:  $\alpha = 0.0094 \pm 0.0010$  at  $k_c = 2\pi$ ,  $T = 0.173$ , against 0.0077 if the friction followed  $\rho_n$ . The arm gives  $0.0096 \pm 0.0018$ , inside the window registered for the temperature law — a confirmation by the rule, although the error is almost twice the one assumed when the window was written, so the arm alone separates the two readings by one standard deviation; the discrimination rests on the comparison across cutoffs.

## 1 Introduction

The mutual friction on a vortex line in a superfluid is parametrised, in the Hall–Vinen form, by  $\alpha$  (along the relative velocity) and  $\alpha'$  (across it). The standard microscopic picture, from phonon and roton scattering, makes the force proportional to the density of excitations and hence  $\alpha \propto \rho_n$  [2, 3]; measurements of the thermal friction on vortices in atomic condensates give values of order  $10^{-2}$  [4, 5], and the only first-principles values for a truncated Gross–Pitaevskii field come from vortex-pair dynamics in the Galerkin-truncated equation [6].

In the preceding paper [1] we measured  $\alpha(T)$  in the *projected* Gross–Pitaevskii equation (PGPE), microcanonical, with no damping term and no noise, with estimators that are identities of the dissipative point-vortex model. The result was  $\alpha = 0.0062(4), 0.014(2), 0.02(1)$  at

$T/T_{\text{BKT}} = 0.14, 0.27, 0.43$ , and the ratio  $\alpha/(\rho_n/\rho) = 0.23, 0.26, 0.22$ . That ratio is the observation this paper tests. We had added an interpretation: for any isotropic bath the drag and the Landau normal density are sums over the same modes with the same weight (Appendix B), so  $\alpha/(\rho_n/\rho)$  is the  $w$ -weighted mean of  $c_g\sigma_{\text{tr}}$  over the modes, which we evaluated at 1.4 healing lengths, “the geometric size of the core”, and we registered the cutoff dependence as the open question: if the coefficient is a property of the vortex it should not depend on the cutoff that defines the bath; the Born law  $\sigma_{\parallel} = \kappa^2 k/8c^2$  [3], extrapolated to the cutoff, predicts  $c \propto k_c$ .

**Registered before any run** (PGPE\_FRICTION\_LAW\_PREREG.md, and the kinetic note of the same day): **FL1**,  $\alpha \propto \rho_n$  at each cutoff; **FL2**, the coefficient  $c = \alpha/(\rho_n/\rho)$  differs between three cutoffs by less than 20 % (the rival: a shift above 40 % growing with  $k_c$ ); **FL3**, a spread of 10–30 %, ordered  $c(2\pi/3) \leq c(\pi) \leq c(2\pi)$ , and the Born scaling 0.67 : 1 : 2 as the sharp alternative.

## 2 Instrument and data

The instrument is that of the preceding paper: PGPE on a periodic box  $L = 64$  (units  $\hbar = m = g = 1$ , mean density 1), projector  $|k| \leq k_c$ , symplectic split-step evolution with energy drift below  $10^{-4}$ ; antiparallel vortex pairs imprinted with the corrected phase-and-Bernoulli-amplitude profile in thermal base states; sub-grid vortex tracking;  $\alpha$  from the energy estimator  $\alpha = -\Delta H/(2 \int \sum |v_s|^2 dt)$  and from a two-coefficient regression of the velocities; block jackknife errors; runs end by annihilation or track loss and a validity cut  $d \geq 4$  is applied. The cutoffs are  $k_c = \pi$  (the published arm, grid  $N = 128$ ),  $k_c = 2\pi/3$  (the same grid, projector at a third of the grid wavenumber) and  $k_c = 2\pi$  (grid  $N = 256$ ,  $dx = \xi/4$ ).

**Base states.** Each base is the equilibrated vortex-free  $e = 0.60$  state re-projected (and Fourier-resampled) under the new cutoff, heated to a target energy by the phonon construction, evolved 1000 time units, and measured over 500 (temperature from the equipartition fit,  $n_s/n = 1 - J_T/J_L$ , raw vortex count); a base is admitted only if the vortex count is below 0.5 (Table 1). The fine-grid base at  $T = 0.217$  (1.32 vortices) failed admission; the second fine temperature is therefore  $T = 0.173$ , not the registered 0.22, and it is the arm that was still running when the temperature law was formulated.

| cutoff            | grid      | $e$   | $T$   | $n_s/n$ | $\rho_n/\rho$ | raw $n_v$ (admitted)         |
|-------------------|-----------|-------|-------|---------|---------------|------------------------------|
| $2\pi/3$          | $N = 128$ | 0.545 | 0.100 | 0.9846  | 0.0154        | 0.00                         |
| $2\pi/3$          | $N = 128$ | 0.60  | 0.216 | 0.9603  | 0.0397        | 0.00                         |
| $\pi$ (published) | $N = 128$ | 0.60  | 0.115 | 0.973   | 0.027         | 0.00                         |
| $\pi$ (published) | $N = 128$ | 0.70  | 0.220 | 0.947   | 0.053         | 0.00                         |
| $2\pi$            | $N = 256$ | 0.95  | 0.127 | 0.9291  | 0.0709        | 0.04                         |
| $2\pi$            | $N = 256$ | 1.10  | 0.173 | 0.9198  | 0.0802        | 0.08                         |
| $2\pi$            | $N = 256$ | 1.25  | 0.217 | 0.9029  | 0.0971        | 1.32 ( <i>not admitted</i> ) |

Table 1: Base states of the campaign. At comparable  $T$  the normal fraction grows steeply with the cutoff ( $\rho_n/\rho = 0.015, 0.027, 0.071$  at  $T = 0.100, 0.115, 0.127$ ): the Landau sum runs over more modes.

**Known answers.** At  $T = 0$  (uniform condensate, antiparallel  $d_0 = 10$ , 400 time units) the pair moves at 0.1065, 0.1068, 0.1065 ( $k_c = \pi, 2\pi/3, 2\pi$ ) and the separation of the four-vortex configuration oscillates between 9.7 and 10.6 with the same phase at all three cutoffs (Fig. 1);  $|\hat{\alpha}| \leq 5 \times 10^{-4}$  and  $1 - \hat{\alpha}' = 0.9986, 0.9995, 0.9991$ . The residual diffusion floor is  $2.9 \times 10^{-5}$

at  $2\pi/3$  against a registered clause of  $2 \times 10^{-5}$ : marginal, reported, and a factor  $10^2$  below the thermal values used below.

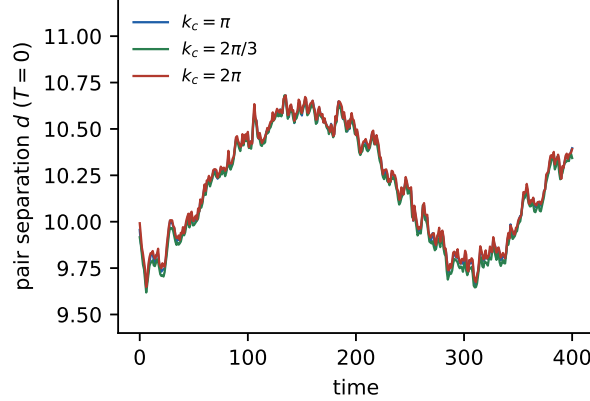


Figure 1: Known answer: the  $T = 0$  pair separation at the three cutoffs. Speed 0.1065/0.1068/0.1065; the curves overlies to 0.5 %.

### 3 Registered predictions and what happened

| arm ( $k_c, T$ ) | runs | $\rho_n/\rho$ | $\alpha$ (energy)   | $c = \alpha/(\rho_n/\rho)$ | $\alpha/T$        | $\alpha'$          |
|------------------|------|---------------|---------------------|----------------------------|-------------------|--------------------|
| $2\pi/3, 0.100$  | 6    | 0.0154        | $0.0049 \pm 0.0008$ | $0.32 \pm 0.05$            | $0.049 \pm 0.008$ | $-0.017 \pm 0.001$ |
| $2\pi/3, 0.216$  | 6    | 0.0397        | $0.0121 \pm 0.0017$ | $0.31 \pm 0.04$            | $0.056 \pm 0.008$ | $-0.031 \pm 0.003$ |
| $\pi, 0.115$     | 8    | 0.0270        | $0.0062 \pm 0.0004$ | $0.229 \pm 0.015$          | $0.054 \pm 0.004$ | $-0.010 \pm 0.002$ |
| $\pi, 0.220$     | 6    | 0.0534        | $0.0138 \pm 0.0023$ | $0.26 \pm 0.04$            | $0.063 \pm 0.010$ | $-0.015 \pm 0.003$ |
| $2\pi, 0.127$    | 6    | 0.0709        | $0.0068 \pm 0.0009$ | $0.096 \pm 0.013$          | $0.053 \pm 0.007$ | $-0.005 \pm 0.002$ |
| $2\pi, 0.173$    | 6    | 0.0802        | $0.0096 \pm 0.0018$ | $0.120 \pm 0.023$          | $0.055 \pm 0.011$ | $-0.004$           |

Table 2: The six arms. Errors are block jackknives over runs;  $\alpha' = 1 - \widehat{(1 - \alpha')}$  is the box-frame value, without the  $T = 0$  pair-size baseline of the preceding paper (all values small and negative; the Iordanskii value would be  $+\rho_n/\rho$ ).

**FL2 fails.** The coefficient is  $0.31 \pm 0.03$ ,  $0.232 \pm 0.014$ ,  $0.102 \pm 0.011$  at  $k_c\xi = 2.1, 3.1, 6.3$  (weighted over the arms at each cutoff):  $(\max - \min)/\text{mean} = 0.98$ , against the registered 0.2. **The rival fails too:**  $c/k_c$  is not constant ( $\chi^2 = 204$  for 2 degrees of freedom) — the coefficient *falls* with the cutoff, the Born law makes it rise. **FL3's expectation fails** (spread 10–30 %). **One common coefficient fails** ( $\chi^2 = 76$  for 5 degrees of freedom). **FL1 holds** at every cutoff where two temperatures exist ( $2\pi/3$ : 0.32, 0.31;  $\pi$ : 0.23, 0.26;  $2\pi$ :  $0.096 \pm 0.013$ ,  $0.12 \pm 0.02$ ) — each cutoff is proportional to its own  $\rho_n$ ; it is the coefficient that depends on the cutoff.

### 4 A temperature law, found after the fact

The comparison of the two panels of Fig. 2 is the finding. At fixed  $T$  the friction hardly moves with the cutoff, while  $\rho_n/(\rho T)$  moves by a factor of three to four (0.15–0.18 at  $2\pi/3$ , 0.24 at  $\pi$ , 0.46–0.56 at  $2\pi$ );  $\alpha/T = 0.0540 \pm 0.0026$  in all six arms ( $\chi^2 = 1.28$  for 5 degrees of freedom, against 76 for a common  $c$ ). In the earlier paper the three temperatures were at one cutoff, where

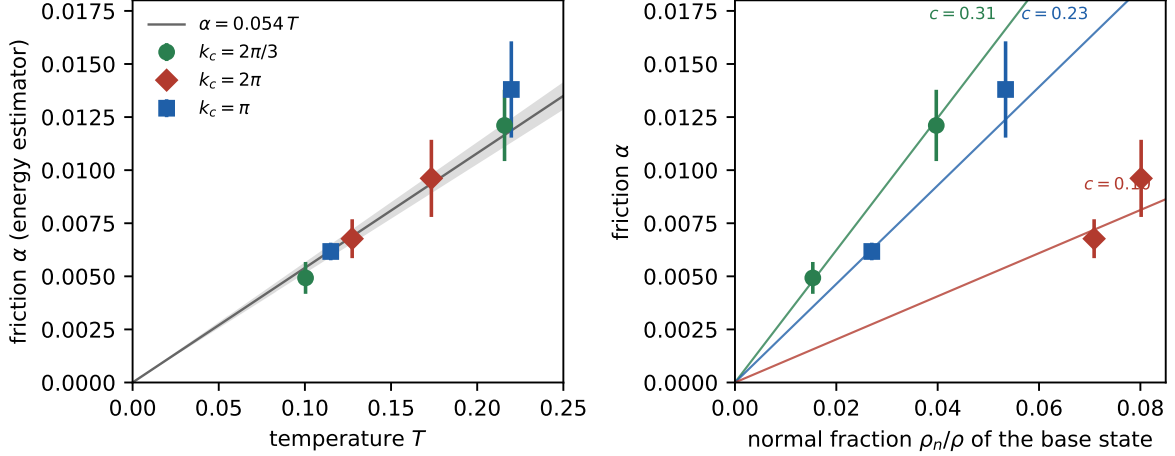


Figure 2: Left: the friction against temperature, for the three cutoffs; the line is  $\alpha = 0.054 T$  (band: one standard error). Right: the same data against the normal fraction of the base state; the three straight lines through the origin have different slopes (Table 2). Open symbols: fewer than six runs.

$\rho_n/\rho \propto T$  (we measured  $\rho_n/(\rho T) = 0.23, 0.24, 0.27$ ), and a proportionality to either could not be told from the other; the cutoff scan breaks the degeneracy.

**Hypothesis H-T** (post hoc; filed as amendment FL-A1 before the last arm was read):  $\alpha = a T$ ,  $a = 0.054 \pm 0.003$ , independent of the cutoff for  $k_c \xi \geq 2$  (a cutoff at 2.1 already carries the whole friction). **Prediction fixed in advance** for the arm running when it was written ( $k_c = 2\pi$ ,  $T = 0.1734$ ,  $\rho_n/\rho = 0.0802$ ): H-T gives  $\alpha_E = 0.0094 \pm 0.0010$ ; the proportionality to  $\rho_n$  at that cutoff's coefficient gives 0.0077; [0.0084, 0.0104] supports H-T, [0.0067, 0.0087] the alternative, the overlap is inconclusive. **Outcome** (read 2026-10-09 by the pre-written rule):  $\alpha_E = 0.0096 \pm 0.0018$  ( $\alpha_{\text{reg}} = 0.0110 \pm 0.0014$ ; six runs, two ended by annihilation, two by track loss),  $c = 0.120 \pm 0.023$ ,  $\alpha/T = 0.055 \pm 0.011$  — **in the window of H-T**, outside that of the  $\rho_n$  law (inclusion by the rule; by itself the arm lies  $0.2\sigma$  from 0.0094 and  $1.0\sigma$  from 0.0077).

**Estimator robustness (found while porting the estimators to Rust).** A re-implementation of the registered estimators in a second language, bit-identical to the Python on four real tracks, was run on the registered synthetic gate over many seeds and showed that the energy estimator of  $\alpha$ , the primary estimator used throughout, is *biased low by the diffusion*: for a true  $\alpha = 0.02$  it returns 0.0200 without diffusion and 0.0183, 0.0163, 0.0156 at  $\eta = 5 \times 10^{-4}, 10^{-3}, 2 \times 10^{-3}$  ( $-8, -18, -22\%$ ), independently of the detection noise, while the regression estimator is unbiased (0.0194); the registered gate passed on one seed. In the data the two estimators differ by 5–21% in that direction. With the regression estimator  $\alpha/T = 0.0598 \pm 0.0023$  ( $\chi^2 = 5.2$  for 5 degrees of freedom;  $0.0540 \pm 0.0026$  with the energy estimator) and the coefficient  $c$  is  $0.355 \pm 0.035$ ,  $0.250 \pm 0.012$ ,  $0.124 \pm 0.011$  at the three cutoffs, a factor 2.9: every conclusion above stands, with absolute values of  $\alpha$  higher by about 11% (the values quoted for  $k_c = \pi$  in the preceding paper are low by 5–20%).

**Why a temperature law is natural, and what it requires.** In a classical field every mode carries the energy  $T$ , so occupations are  $n_k = T/\varepsilon_k$  and anything linear in the occupation is linear in  $T$ . The kinetic form  $D = \frac{1}{2} \int \frac{d^2 k}{(2\pi)^2} \frac{T}{\varepsilon_k^2} k^2 c_g \sigma_{\text{tr}}$  with the Bogoliubov energy  $\varepsilon_k = \sqrt{k^2 + k^4/4}$  is

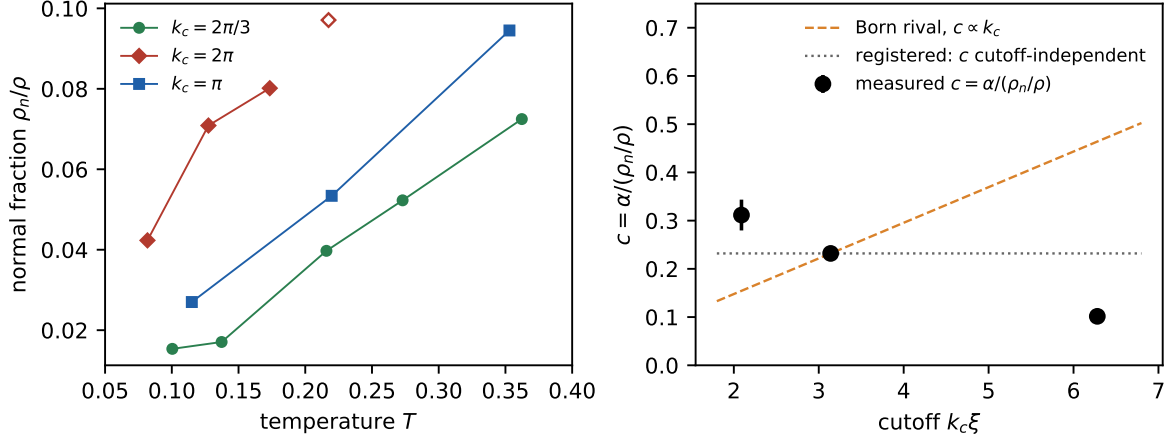


Figure 3: Left: the normal fraction against temperature at the three cutoffs (open: not admitted). Right: the coefficient  $c$  against the cutoff: the registered (constant) and rival ( $\propto k_c$ ) predictions fail.

then  $T$  times a function of the cutoff; for  $k\xi \gg 1$  ( $\varepsilon \simeq k^2/2$ ,  $c_g \simeq k$ ) the integrand is  $\propto \sigma_{\text{tr}}(k)$ , so a drag that does not depend on the cutoff beyond  $k_c\xi \simeq 2$  requires  $\int^\infty \sigma_{\text{tr}} dk$  to converge — a transport cross-section falling faster than  $1/k$  — whereas the same weights make  $\rho_n$  grow logarithmically with the cutoff. This is a constraint on the scatterer if the independent-mode picture holds; the long-wavelength Born cross-section  $\sigma_{\parallel} = \kappa^2 k/8c^2$  grows linearly with  $k$  and is valid for fewer than 3% of the modes of a cutoff  $\pi$  [3], so it cannot be extrapolated. The direct measurement of  $\sigma_{\text{tr}}(k)$  — a monochromatic Bogoliubov wave scattered by one vortex at  $T = 0$  — would settle it and is the obvious next experiment.

## 5 What this corrects

1. The statement “ $\alpha \simeq 0.24 \rho_n/\rho$ ” of the preceding paper [1] holds at  $k_c = \pi$  only; the relation that survives the cutoff scan is  $\alpha = 0.054 T$  (confirmed on the last arm by the registered rule).
2. The reading of 0.24 as a mode-averaged transport cross-section of the vortex,  $\langle c_g \sigma_{\text{tr}} \rangle = 1.4 \xi$ , “the geometric size of the core”, is withdrawn: the same quantity is 1.9, 1.4, 0.59  $\xi$  at the three cutoffs. The identity of Appendix B stands; its coefficient is a property of the bath’s mode set, not of the vortex.
3. The recommendation to compare with quantum gases at equal  $\rho_n/\rho$  rather than equal  $T/T_c$  is withdrawn in the same sense: in this model  $\rho_n/\rho$  depends on the cutoff,  $\alpha$  does not. (A comparison with experiment needs a map from the classical-field temperature to the physical one, and that map itself depends on the cutoff; the claim here is internal to the model.)

## 6 Limits

One box ( $L = 64$ ), one interaction strength ( $mg = 1$ ), pairs of two sizes and two or three temperatures per cutoff; the two new cutoffs at one and two temperatures; the dependence on the pair size ( $\alpha$  larger for  $d_0 = 12$  than for  $d_0 = 8$  by 38–85% at the new cutoffs, not at  $\pi$ ) and the small negative transverse coefficient, whose magnitude grows toward the coarse cutoff ( $\alpha' = -0.005$  at  $2\pi$ ,  $-0.010$  and  $-0.015$  at  $\pi$ ,  $-0.017$  and  $-0.031$  at  $2\pi/3$ ; always far from the Iordanskii value) are reported and unexplained — either a bias of the energy estimator at the

coarse cutoff or a finite-pair effect. The cutoff  $2\pi/3$  shares its grid with  $\pi$  and the cutoff  $2\pi$  has a finer grid: a lattice-spacing effect is not excluded by this design, though the  $T = 0$  pair is insensitive to it to 0.3 %. The energy estimator of  $\alpha$  is biased low by diffusion (above), so the absolute coefficient is 0.054–0.060; the law is estimator-independent in its form. The temperature law was formulated after five arms were read and confirmed by the rule on the sixth, whose error is large; it is a well-supported empirical relation over  $T = 0.10$ – $0.22$  and three cutoffs, not a derived one, and it has not been tested above  $T = 0.22$  (where thermal vortices enter) or at other box sizes or interaction strengths.

## 7 Conclusion

A pre-registered test failed in the way that matters: the coefficient between the friction and the normal density is not a property of the vortex, and neither the registered cutoff-independence nor its Born rival describes it. What the test found instead is a friction that follows the temperature and ignores the cutoff over a factor three, which is a stricter statement than the one it replaces, and a correction of the preceding paper’s reading, made here rather than left to a reader.

**Data and code availability.** Pre-registrations, results logs, the ledger (claims 089–098), analysis scripts, the Lean sources and the figure scripts are in the repository <https://github.com/xaviercallens/SocrateAI-Scientific-QuantumFluids> (scripts as of commit 2ff887a4f57a); the numerical record up to the preceding paper is archived in [7]. The pure-Rust re-implementation of the engine and estimators that exposed the estimator bias is in the rusty-SUNDIALS crate `qf-pgpe`.

## A Registered items and outcomes

| item                | registered (before any run)                          | outcome                                                                                                  |
|---------------------|------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| FL1                 | $\alpha \propto \rho_n$ at each cutoff               | holds at all three cutoffs (each to its own $\rho_n$ )                                                   |
| FL2                 | $c$ within 20 % across cutoffs                       | fails (spread 0.98)                                                                                      |
| Born rival          | $c \propto k_c$                                      | rejected ( $\chi^2 = 204/2$ )                                                                            |
| FL3                 | spread 10–30 %, $c(2\pi/3) \leq c(\pi) \leq c(2\pi)$ | fails (ordering reversed)                                                                                |
| $T = 0$ control     | speed within 4 %, $ \alpha  \leq 10^{-3}$            | passes at all cutoffs; $\eta$ floor marginal at $2\pi/3$                                                 |
| G0 energy criterion | $\alpha_E$ within 15 % (one numpy seed)              | passes on 2 of 12 seeds: biased low by diffusion (regression unbiased)                                   |
| base admission      | raw $n_v < 0.5$                                      | fine $T = 0.217$ base fails; replaced by $T = 0.173$ (admitted, $n_v = 0.08$ )                           |
| H-T                 | post hoc, FL-A1                                      | consistent ( $\chi^2 = 1.28/5$ ); registered window for the last arm met (0.0096 in $[0.0084, 0.0104]$ ) |

## B Lean

**FrictionKinetic:** for a finite set of modes with nonnegative weights  $w_k$  and drag factors  $f_k = c_g \sigma_{\text{tr}}$ ,  $D = \sum w_k f_k$  and  $\rho_n = \sum w_k$  give  $\alpha/(\rho_n/\rho) = \frac{\rho}{\rho_s \kappa} \cdot \sum w_k f_k / \sum w_k$  (an identity), the weighted mean lies between  $\min f$  and  $\max f$ , and for equal weights (Rayleigh–Jeans with a phonon dispersion) it is the arithmetic mean. Nothing about the value of  $\sigma_{\text{tr}}$  or its dependence on  $k$  is asserted; that is the measurement.

## C Code

The analysis (`exploration/pgpe/analyze_friction_law.py`) applies the registered estimators to every arm and writes the verdict file; the core of the cutoff comparison is

```
def wmean(v, s):
    v, s = np.array(v), np.array(s); w = 1 / s ** 2; m = (w * v).sum() / w.sum()
    return float(m), float(1 / np.sqrt(w.sum())), float(((v - m) ** 2 * w).sum()) # mean, se, chi2

c_by_cutoff = {k: wmean([x["c_energy"] for x in v], [x["c_energy_se"] for x in v]) for k, v in
    arms_by_cutoff.items()}
spread = (max(m for m, _, _ in c_by_cutoff.values()) - min(m for m, _, _ in c_by_cutoff.values())) /
    mean_of_means # FL2: < 0.20
m_T, se_T, chi2_T = wmean([a["alpha_energy_over_T"] for a in arms], [a["alpha_energy_over_T_se"] for a in
    arms]) # H-T
```

Data, scripts, pre-registrations and the ledger: [7].

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